

# Ayush Chachan

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## EDUCATION

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**Indian Institute of Technology, Delhi**

July 2016 – June 2022

*Dual Degree (B. Tech and M. Tech) in Biochemical Engineering and Biotechnology*

*CGPA - 8.905/10.000*

## THINGS I STUDY

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- **Books** – Computer Systems: A Programmer's Perspective, Operating System Concepts, Designing Data-Intensive Applications, Database Management Systems, Introduction to Algorithms, Computer Networking: A Top-Down Approach, Operating Systems Three Easy Pieces, Mathematics for Computer Science

## EXPERIENCE

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**Fluxx – Secure Analytics & Engagement Platform** | *Founding Backend Engineer*

Feb 2025 – Present

- **Event Ingestion & Data Pipeline**
  - \* Built a high-throughput Kafka-based event ingestion pipeline consuming SDK-emitted data over a custom gRPC binary protocol.
  - \* Designed a compact binary event encoding format tailored for analytics workloads, reducing serialization overhead versus JSON.
  - \* Implemented batch ingestion into BigQuery to optimize write throughput and downstream query performance.
- **Funnel Analytics Engine & DSL**
  - \* Implemented a user-centric funnel analytics engine operating on aggregated user-event arrays instead of self-join-heavy event-level queries.
  - \* Reduced funnel evaluation complexity from  $O(n^2)$  to  $O(n)$ , achieving **80% lower compute cost** than standard engagement platforms.
  - \* Designed a declarative Funnel DSL enabling configuration-driven execution with support for occurrence constraints, hold properties, uniqueness, and multi-attribute splits.
  - \* Implemented a logical-to-physical execution planning layer to optimize funnel evaluation strategies based on query shape and constraints.
- **Encrypted Analytics & Secure Computation**
  - \* Designed an encrypted analytics execution model with event data encrypted at rest, in transit, and during intermediate processing.
  - \* Implemented envelope encryption using AES-GCM with asymmetric public-key protection (RSA/ECC) for key exchange.
  - \* Built analytics primitives capable of operating on encrypted attributes, supporting equality, inequality, and ordered comparisons.
- **Scalable User Segmentation**
  - \* Designed a hash-partitioned user segmentation system using deterministic sharding ( $hash(user\_id) \bmod N$ ).
  - \* Used column-oriented storage layouts to ensure predictable performance under high-cardinality attributes.

**UPSC Civil Services Preparation** |

April 2023 – Jan 2025

- Prepared for UPSC CSE with **Mathematics Optional**, involving sustained self-study and structured learning.

**Helloverify India Private Limited** | *Software Engineer*

July 2022 – Mar 2023

- Built scalable REST APIs using **Django REST Framework**, improving client onboarding speed and efficiency.
- Engineered **Selenium**-based web scrapers to automate data extraction, reducing the time by **82% per request**.
- Implemented **Celery**-based **schedulers** for asynchronous email and phone alerts, ensuring timely client updates.

**Indian Institute of Technology, Delhi** | *Graduate Researcher (M.Tech Thesis)*

May 2021 – June 2022

- Designed **statistical algorithms** for bioreactor data validation using mass balances and regression models.
- Applied **error propagation analysis** and consistency testing to identify anomalous and unreliable process data.
- Developed Python routines for **least squares estimation** and **chi-square testing** in over-determined systems.

## SCHOLASTIC ACHIEVEMENTS

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**IIT Delhi BOSS Award (2022)** - Conferred for best experimental and design-based research in the **M.Tech project**.

## APPLICATION PROJECTS

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### **Kafka-Like Log-Structured Messaging System** | *Java, Distributed Systems*

- Built from scratch a **high-throughput append-only log** inspired by Kafka for large-scale event ingestion.
- Implemented **log segmentation**, **offset-based indexing**, and **sequential disk I/O** to optimize producer appends and consumer reads.
- Modeled core messaging primitives including **topics**, **partitions**, **offsets**, and **immutable log segments** without using Kafka libraries.

### **Mini Spring MVC Framework with Custom HTTP Server** | *Java, Networking, Concurrency*

- Implemented a **custom HTTP server from scratch**, parsing raw TCP byte streams and handling core **HTTP/1.1 request semantics**.
- Designed a **thread-pool ExecutorService** to enforce thread limiting and backpressure.
- Built a lightweight **IoC container** with custom annotations for component scanning, bean registration, and dependency injection.
- Implemented a **Dispatcher and Controller Method Registry** enabling Spring-like MVC request pipeline.

## CORE CS PROJECTS

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### **Unix-like Shell Implementation** | *C, UNIX System Calls*

- Developed a Unix-style shell using **system calls (fork, exec, wait)** with **zombie process reaping** and robust error handling.
- Implemented **background (&)**, **serial (&&)**, and **parallel (&&&)** execution with custom **signal handlers**.

### **Monte Carlo Simulation** | *Disjoint Sets, Union Find, Java*

- Modeled 2-D lattices using **graphs** and estimated percolation thresholds via **Monte Carlo simulation**.
- Optimized runtime using the **Disjoint Set (Union-Find)** data structure with path compression.

### **K-Dimensional Tree** | *Java, BST, K-Nearest Neighbour Search*

- Implemented a **2D KD-Tree** for indexing points in a **K-dimensional space**.
- Supported Insert, Search, **Range Search**, and **Nearest Neighbour** queries in  $O(\log^2 n)$  time.

## TECHNICAL SKILLS

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**Programming Languages:** Java, Python, SQL, Java Script, C/C++, Bash (Shell Scripting).

**Frameworks & Libraries:** Spring Boot, Hibernate/JPA, Django REST Framework, Celery

**Databases & Storage:** MySQL, PostgreSQL, Redis.

**Web & API Development:** REST APIs, Microservices, JSON, HTML5, CSS3.

**Developer Tools:** Git, Maven, Amazon Q, Bash, .

**Data Analysis:** Jupyter, NumPy, Pandas, Excel (Data Visualization).